

```

MISCLOSE;MISCE;MISCN;MISCH;IO;SNUMBERS
MISCE+COORDS[IF[I],1]+N[I]-COORDS[IT[I],1]
MISCN+COORDS[IF[I],2]+N[I]-COORDS[IT[I],2]
MISCH+COORDS[IF[I],3]+N[I]-COORDS[IT[I],3]
→(OINDEX[IF[I]]=OINDEX[IT[I]])/SAMEORIGIN
→(OCURR≥OINDEX[IF[I]][OINDEX[IT[I]])/NOTNEW
CHEAD 3
IO←IF[I]
→(OINDEX[IF[I]]>OINDEX[IT[I]])/IOISOK
IO←IT[I]
IOISOK:OCURR←OINDEX[IO]
ORIGINOUT IO
→CONNECTOUT
NOTNEW:CHEAD 2
CONNECTOUT:SNUMBERS←(4 0 7OINDEX[IF[I]]), ' AND ',(4 0 7OINDEX[IT[I]])
REC6←(15 0 ' '), 'SURVEYS', SNUMBERS, ' CONNECT'
LINECOUNT←LINECOUNT+1
→PRINT
SAMEORIGIN:CHEAD 1
PRINT:MISCLOSUREOUT

```

PRINTSTATISTICS

→0

STOREDATABASE

CALIBOUT

CALIBOUT

LINECOUNT←LINECOUNT+1

VHEAD

REC6←' CALIBRATION :',(16p' '), (8 1 7DEC), 8 1 7CLINO

MISCLOSE

```
A      SURESET IS NEEDED TO CLOSE SHARING WITH THE FILE-PROCESSOR
A      BEFORE SU CAN BE RUN AGAIN IN THE SAME APL SESSION.
A      SURESET IS PRONOUNCED  S-U-RESET.
```

VECTOR

```
VECTOR;BEAR,ELEV
BEAR←DATA[2]×RAD
ELEV←DATA[3]×RAD
LENGTH←LENGTH+DATA[1]+DATA[4]
→FIRSTTIME/FIRSTLEG
IF←IF,LIST FROM
IT←IT,LIST TO
DIST←DIST,DATA[1]×20ELEV
E←E,DIST[ρDIST]×10BEAR
N←N,DIST[ρDIST]×20BEAR
H←H,DATA[4]+DATA[1]×10ELEV
PLANLN←PLANLN+DIST[ρDIST]
→PRINT
```

```
A
FIRSTLEG:IF←1ρLIST FROM
IT←1ρLIST TO
DIST←1ρDATA[1]×20ELEV
E←DIST×10BEAR
N←DIST×20BEAR
H←1ρDATA[4]+DATA[1]×10ELEV
PLANLN←DIST
FIRSTTIME←0
PRINT:OUTVECTOR
```

SU

```
SU
SUINITIALISE
SETUPFILES
FIRSTREAD
READLOOP:ENDOFDATA←GETDATA
→ENDOFDATA/ALLDATAREAD
VECTOR
```

CHL1000
INPUT←REC5
STA← 1 6 p6↑INPUT
OINDEX←1p1
COORDS←ORIGINPUT INPUT

T

TIME←T
TIME←(10p' '), (TOTS[4]), ': ', TOTS[5]

SURESET

SURESET

```
DATA[2]*DATA[2]-DEC
DATA[3]*DATA[3]-CLINO
COMMENT+36*INPUT
+ (~INFEET)/0
DATA[1 4]*DATA[1 4]*0.3048
```

VECOUNT

```
OUTPUT+VECOUNT 1
OUTPUT+' ',STAC[IF[I]],', ' ',STAC[IT[I]],', 9 3 *(E[I],N[I],W[I],DIST[I])
```

UNDEFINEDOUT

```
UNDEFINEDOUT
CHEAD 1
REC6+(VECOUNT 1), ' UNDEFINED'
LINECOUNT+LINECOUNT+1
```

SUINITIALISE

```
SUINITIALISE
VERSHN+'SU-AFL 1.00'
ZER0ES+4*0
BLANKS+' '
OCOUNT+1
INFEET+0
RAD+0+180
LENGTH+0
FIRSTTIME+1
```

FIRSTREAD

```
FIRSTREAD
INPUT+REC5
DEC+FLD 6*INPUT
CLINO+FLD 6*12*INPUT
TITLE+12*INPUT
```

```
SETUPFILES;FILENAME  
NAME←9p(PROMPT 'ENTER NAME OF SURVEY : '), 'Z'  
A  
FILENAME←NAME, ' DATA'  
RESULT5←FILE 5  
FILENAME←NAME, ' OUT A0 (RECFM U BLOCK 134'  
RESULT6←FILE 6
```

SURVEYDATA

```
SURVEYDATA  
TO←6↓12↑INPUT  
DATA←(FLD 6↑12↓INPUT),(FLD 6↑18↓INPUT),(FLD 6↑24↓INPUT),(FLD 6↑30↓INPUT)
```

```
COORDS[IUNDEFINED;3]+COORDS[IDDEFINED;3]+A×H[I]  
OINDEX[IUNDEFINED]+OINDEX[IDDEFINED]  
→(OINDEX[IDDEFINED]≤OCURR)/NONEWORIGIN  
OCURR←OINDEX[IDDEFINED]  
CHEAD 2  
ORIGINOUT IDDEFINED  
→PRINT  
NONEWORIGIN:CHEAD 1  
PRINT:STATIONOUT
```

PRINTSTATISTICS

```
PRINTSTATISTICS  
REC6←'1 TOTAL LENGTH OF SURVEY :', 9 2 √LENGTH  
REC6←' PLAN LENGTH :', 9 2 √PLANLN
```

GETDATA

```
ENDOFDATA←GETDATA;OPTIONS  
OPTIONS← 5 6 p'END CHANGEFEET METRICORIGIN'  
READ:INPUT←REC5  
→(0=pINPUT)/LAST  
FROM←6↑INPUT  
→(^(5 6 pFROM)=OPTIONS)/(LAST,CHANGE,FEET,METRIC,NEWORG)  
ENDOFDATA←0  
SURVEYDATA  
→0  
CHANGE:DEC←FLD 6↑12↓INPUT  
CLIND←FLD 6↑18↓INPUT  
CALIBOUT  
→READ  
FEET:INFEET←1  
→READ  
METRIC:INFEET←0  
→READ  
NEWORG:OINDEX[LIST 6↑6↓INPUT]+OCOUNT←OCOUNT+1  
COORDS[pOINDEX;]←ORIGINPUT INPUT  
→READ  
LAST:ENDOFDATA←1
```

```
1.3 * 1  
→(Λ/STA=LpST)/0  
IS←2  
→APPEND  
LONGLIST:IS+(Λ/STA=LpST)11  
→(IS≤L[1])/0  
APPEND:STA←STA,[1] ST  
OINDEX←OINDEX,0  
COORDS←COORDS,[1] 0 0 0
```

NEXTSTATION

```
NEXTSTATION  
COORDS[IUNDEFINED;1]←COORDS[IDDEFINED;1]+A×E[1]  
COORDS[IUNDEFINED;2]←COORDS[IDDEFINED;2]+A×N[1]
```

COORDINATES

COORDINATES;IMAX,I,A,IDEFINED,IUNDEFINED

OCURR←0

IMAX←p IF

I←1

LOOP:→(OINDEX[IF[I]]>0)/FSIGHT

→(OINDEX[IT[I]]>0)/BSIGHT

A ISOLATED LEG

UNDEFINEDOUT

→LOOPEND

A

FSIGHT:→(OINDEX[IT[I]]>0)/CLOSURE

A FORWARD SIGHTING LEG

A←1

IDEFINED←IF[I]

IUNDEFINED←IT[I]

→CALCULATE

A BACK SIGHTING LEG

BSIGHT:A←1

IDEFINED←IT[I]

IUNDEFINED←IF[I]

A

CALCULATE:NEXTSTATION

→LOOPEND

CLOSURE:MISCLOSE

LOOPEND:LINECOUNT←LINECOUNT+1

→(IMAX≥I+I+1)/LOOP

ORIGINPUT

XYZ←ORIGINPUT DATA

XYZ← 1 3 p(FLD 12↑12↓DATA),(FLD 12↑24↓DATA),(FLD 12↑36↓DATA)

LIST

IS←LIST ST;L

```
REC6←PRNT1,STHL1/1/ 10 3 7COUNT+1/1/1  
LINECOUNT←LINECOUNT+1
```

PROMPT

```
Z←PROMPT MSG  
M←MSG  
Z←(pMSG)↓M
```

MISCLOSUREOUT

```
MISCLOSUREOUT  
REC6←(VEOUT 1), ' MISCLOSES', 10 3 7(MISCE,MISCN,MISCH)  
LINECOUNT←LINECOUNT+1
```

```
DATA←AINPUT  
→0  
ALLBLANK:DATA←0
```

CHEAD

```
CHEAD I;HED1;HED2;K  
K←0  
LOOP:K←K+1  
→(K>I)/0  
→(1≠40|LINECOUNT+K)/LOOP  
HED1←' FROM TO EAST NORTH HEIGHT'  
HED2←' PLAN STATION EAST NORTH HEIGHT'  
REC6←'1 ',VERSHN,(20p' '),TITLE  
REC6←' '  
REC6←(22p' '), 'VECTOR', (45p' '), 'POSITION'  
REC6←' ', (49p' -'), ' ', 38p' -'  
REC6←HED1,HED2  
REC6←' '  
LINECOUNT←LINECOUNT+K-1
```

DISPLAYFN

```
DISPLAYFN  
X←ONL 3  
I←(pX)[1],5  
Z←I p'DCR ''  
K←1  
LOOP:' '  
X[K;]  
' '  
A(Z,X)[K;], ''  
' '  
K←K+1  
→(K≤I)/LOOP
```

ORIGINOUT

```
100 DSVO 'CMS'  
CMS←'FILEDEF ',APLN,' CLEAR'  
CMS←'FILEDEF ',APLN,' DISK ',FILENAME  
ARECN, '←'',APLN,' (370''  
ACTLN, '←'',APLN,' (CTL''  
RESULT←111 DSVO 2 4 ρRECN,CTLN  
A'RCODE←',RECN  
A'CCODE←',CTLN  
ACTLN, '← 1 0 1'
```

FLD

```
DATA←FLD INPUT  
→(^(ρINPUT)ρ' '=INPUT)/ALLBLANK
```

DISPLAYFN

OUTVECTOR

```
OUTVECTOR,I,PART1
LINECOUNT←LINECOUNT+1
VHEAD
PART1←' ',FROM,' ',TO,' ',(8 2 ¶DATA[1]),(8 1 ¶DATA[2 3])
PART1←PART1,(8 2 ¶DATA[4]),(10p' ')
I←pE
→(I<1.5)/SCALAR
REC6←PART1, 8 3 ¶(E[I],N[I],H[I],DIST[I])
→0
SCALAR:REC6←PART1, 8 3 ¶(E,N,H,DIST)
```

STATIONOUT

```
STATIONOUT
REC6←(VEOUT I),(7p' '),STA[IUNDEFINED;], 10 3 ¶COORDS[IUNDEFINED;]
LINECOUNT←LINECOUNT+1
```

VHEAD

```
VHEAD;HED1;HED2
→(1≠40|LINECOUNT)/0
HED1←' FROM TO TAPE COMPASS CLINO PLUMB'
HED2←' PLAN EAST NORTH HEIGHT'
REC6←'1',(15p' '),VERSHN,' ':',TITLE
REC6←' '
REC6←(23p' '), 'SURVEY DATA', (31p' '), 'COMPUTED VECTOR'
REC6←(23p' '), (32p' -'), (10p' '), 32p' -'
REC6←' '
REC6←HED1,HED2
REC6←' '

```

FILE

RESULT←FILE N;CMS;REC�;CTLN;APLN

SU-APL.

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